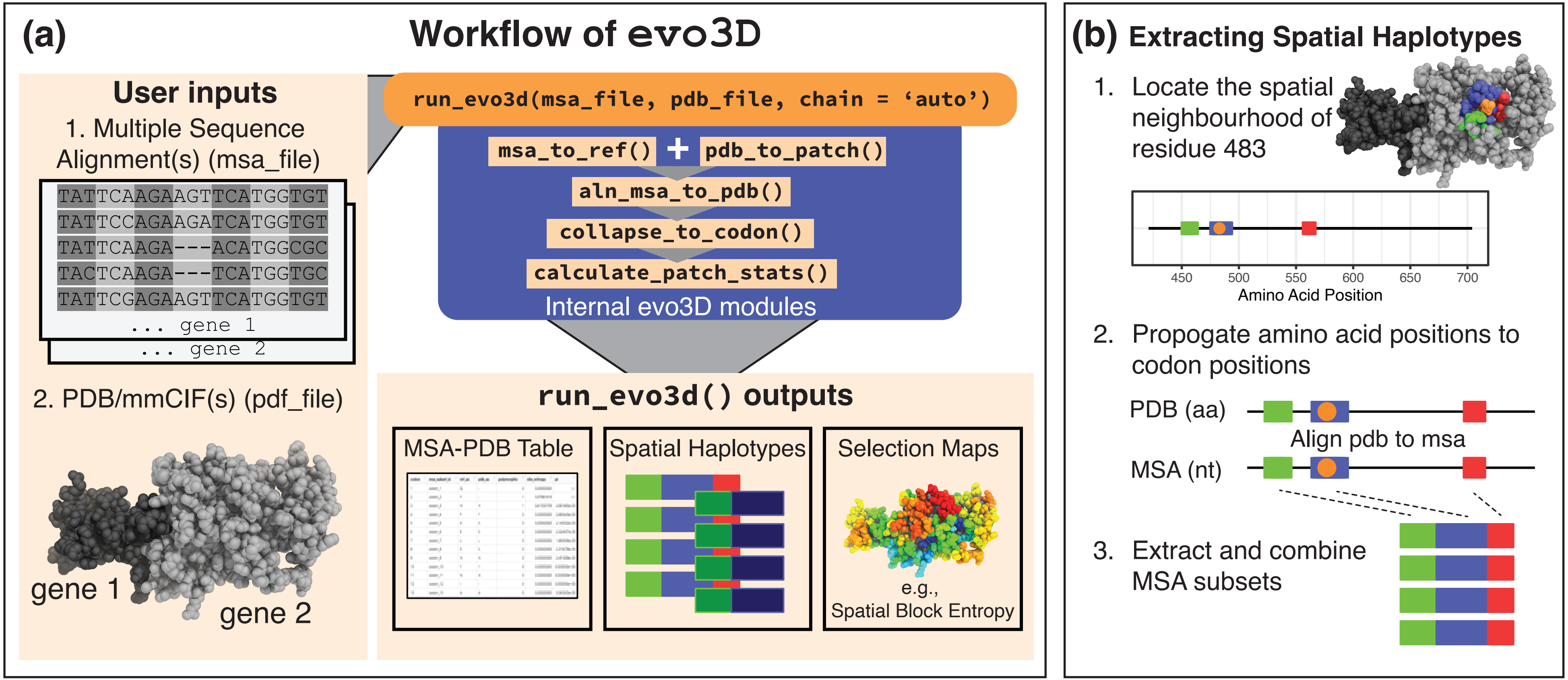
(a)
Workflow of evo3D
User inputs
1. Multiple Sequence Alignment(s) (msa_file)
TATTCAAGAAGTTCATGGTGT
TATTCCAGAAGATCATGGTGT
TATTCAAGA---ACATGGCGC
TACTCAAGA---TCATGGTGC
TATTCGAGAAGTTCATGGTGT
... gene 1
... gene 2
2. PDB/mmCIF(s) (pdf_file)
gene 1
gene 2
run_evo3d(msa_file, pdb_file, chain = 'auto')
msa_to_ref()
pdb_to_patch()
aln_msa_to_pdb()
collapse_to_codon()
calculate_patch_stats()
Internal evo3D modules
run_evo3d() outputs
MSA-PDB Table
Spatial Haplotypes
Selection Maps
e.g.,
Spatial Block Entropy
(b) Extracting Spatial Haplotypes
1. Locate the spatial neighbourhood of residue 483
450
500
550
600
650
700
Amino Acid Position
2. Propogate amino acid positions to codon positions
PDB (aa)
Align pdb to msa
MSA (nt)
3. Extract and combine MSA subsets